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AI25BTECH11034 - Sujal Chauhan

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Question

Let $\mathbf{M}$ be a 3×3 matrix such that

$$\mathbf{M} \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix}$$

and suppose that

$$\mathbf{M}^3 \begin{pmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

for some $\alpha, \beta, \gamma \in \mathbb{R}$. Then $|\alpha|$ is equal to _____.

Solution

Let $\mathbf{M}$ be a 3×3 matrix such that

$$\mathbf{M} \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix} \quad (1)$$

We observe that

$$\begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix} = -3 \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \quad (2)$$

Hence, $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ is an eigenvector of $\mathbf{M}$ with eigenvalue $\lambda = -3$.

Solution

Now consider

$$\mathbf{M}^3 \begin{pmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \quad (3)$$

Note that

$$\begin{pmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \quad (4)$$

which is a scalar multiple of the same eigenvector. Therefore, it also corresponds to the same eigenvalue $\lambda = -3$.

Solution

Applying the matrix $\mathbf{M}$ three times will scale the eigenvector by $\lambda^3 = (-3)^3 = -27$. Hence,

$$\mathbf{M}^3 \begin{pmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{pmatrix} = (-27) \begin{pmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} -27 \\ \frac{27}{2} \\ 0 \end{pmatrix} \quad (5)$$

Thus,

$$\boxed{|\alpha| = 27.} \quad (6)$$